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RESEARCH ARTICLE

Optimal Activity Recognition Framework Based on Improvement of Regularized Neighborhood Component Analysis (RNCA)

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ABSTRACT This study aims to investigate the alternative model structure based on a feature selection algorithm on multiple features-framework of human activity recognition (HAR) via wearable sensor-based modality. Neighborhood component analysis (NCA) is a linear transformation that maximizes the accuracy of specified classification events used as the benchmark for the proposed algorithm. Also, the effect of different combinations of sensor configurations of two, three, and all four sensors on the performance of the developed model was studied. The effectiveness and shortcomings of best sensor configuration were highlighted. Results were compared between different sensor configurations and benchmark HAR dataset. To maximize the regularization of NCA, fine-tuning the algorithm to maximize relevance and minimize redundancy (MRMR) was proposed. Results demonstrated that RNCA-MRMR could establish an efficient algorithm that can satisfy the model validation tests with significant advantages over feature number and predictive accuracy at 93.5%, 93.7%, and 94.5% for two, three, and all four sensors respectively. Furthermore, the adaptability of RNCA-MRMR to different data characteristics has ensured an optimal and task-specific representation of the data. In essence, the combined strength of RNCA and MRMR provides a versatile and effective approach for extracting meaningful features and enhancing the overall performance of machine learning models.

INDEX TERMS Accelerometer, activity recognition, feature selection, gyroscope, machine learning, wearable sensor.

I. INTRODUCTION

Human activity recognition (HAR) is an essential field in the field of artificial intelligence (AI) and sensing technologies, with significant importance across various domains [1]. This technology automatically identifies and classifies human activities based on sensor data or visual inputs, and its applications are profound. For instance, wearable devices equipped with accelerometers and gyroscopes enable continuous monitoring of patients' activities and health conditions, aiding in early disorder detection and rehabilitation tracking. HAR is also employed in the sports and fitness industry for performance analysis, training optimization, and injury prevention, providing valuable insights into the biomechanics and movement patterns of athletes. In addition, smart homes benefit from HAR, as it automates

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tasks, enhances security, and increases convenience through activity-based control systems. The surveillance and security sector can also leverage HAR's ability to detect suspicious behaviours or unauthorized access, improving overall surveillance efficiency. Moreover, HAR is integral to the field of Human-Computer Interaction (HCI), enabling gesture recognition, sign language interpretation, and enhancing user experiences in virtual and augmented reality applications. Therefore, the diverse applications of HAR underscore its significance in shaping the future of technology and human interaction.

Although HAR presents immense potential, it faces several challenges and processes. Such as collection of data for labelled datasets can be resource-intensive, raising concerns regarding privacy and ethical considerations [2]. Secondly, the variability and complexity of human activities in different contexts pose challenges in creating robust recognition models [2]. Furthermore, the real-time processing requirements, especially in applications like security and healthcare, necessitate efficient algorithms and low-latency hardware [2]. Ensuring generalization to new users, activities, and dynamic environments remains an ongoing challenge. Scalability is another concern, as developing systems that accommodate a broad spectrum of activities and scenarios is a complex endeavour [2]. The future of HAR is full of promise, with several exciting developments on the horizon. One of the most significant advancements that have been anticipated is the increasing importance of explainable AI models. These models are expected to enable systems to provide interpretable explanations for their decisions, which is critical in building user trust and adoption. In particular, the application of this technology in healthcare and security sectors is poised to transform the industry. Another crucial aspect of future of HAR is the role that Edge computing is set to play. By incorporating various sensors such as environmental and physiological data, we can enable more comprehensive activity recognition and context awareness. Finally, lifelong learning approaches that allow systems to continually adapt and learn from new data and user interactions will enhance their generalization capabilities. In summary, the journey of HAR continues to be marked by significant advancements that are shaping the landscape of technology and its profound impact on human activities and interactions. Following are the contributions the study:

- Improved model performance by selecting relevant features: The improvement of feature selection has resulted in increased accuracy, reliability, and reduced complexity of the model, leading to better generalization and faster computation.
- Increased interpretability of the model: The selection of key features has facilitated the interpretability of the classification models. By identifying which features play a vital role in distinguishing different activities, valuable insights into the underlying patterns and characteristics of human activity have been gained.
- Enhanced robustness. Improves the robustness of the classification model by minimizing the impact of irrelevant information, thereby improving the model's ability to handle variations of human activity datasets.

II. LITERATURE REVIEW

A. AUTOMATED HAR USING WEARABLE SENSOR-BASED MODALITY

Wearable sensors integrated into the user's body while performing activities are referred to as wearable sensor-based modality. Existing works on wearable sensors for human activity recognition (HAR) have been reviewed in several studies [2], [3], [4], [5], [6]. The development of miniature sensors has enabled users to conveniently interact with devices as part of their daily lives. The recognition of human activities has become an area of high interest, predominantly for monitoring ADLs [7], [8], [9], rehabilitation [10], [11], and sports analysis [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39]. However, there are still challenges that motivate the development of new techniques to improve accuracy under more realistic conditions. These challenges include constructing a portable and inexpensive data acquisition system that is unobtrusive, collecting data under realistic conditions [2], [4], [6], implementing mobile devices that meet energy and processing requirements, selecting attributes to be measured, designing feature extraction and inference methods [6], and ensuring flexibility to support new users without the need for re-training the system. To minimize obtrusiveness during use, sensor devices are often integrated into platforms such as smartphones [6], smartwatches and inertial units.

Wearable sensors offer the advantage of functioning in a relatively large space when the wearer is in motion [40]. Modern technologies such as smartphones, smartwatches, and inertial sensor are the mainstream products embedded in wearable technologies in HAR. These sensors differ from commonly used industrial sensors, as they are designed to meet specific requirements such as high integration density, small size [2], [40], low power consumption [2], and high measurement accuracy [2]. They are physically integrated into a small device that can be conveniently attached to the user's body parts [40]. Wearable sensors can be comprised of inertial sensors, physical health sensors, environmental sensors, cameras, and microphones. Accelerometers, gyroscopes, and magnetometers are the most popular motion-based inertial sensors that have been successfully applied in wearable sensor-based modalities [2]. They detect and measure acceleration, angular velocity, magnetic fields, tilt, shock, vibration, rotation, and multiple degrees-of-freedom motion. These observations vary sensitively with a wearer's movement or body postures, thereby delivering rich motion-caused information. Accelerometers, for example, are adept at identifying physical activities such as walking, jogging, ascending, and descending stairs, sitting,

and standing [2]. Different models such as the Reduced Kernel Extreme Learning or the use of an accelerometer, a magnetometer, and a gyroscope built into a smartphone have been developed for patient activity recognition. While inertial sensors still have limitations such as calibration for effective measurements [2], battery life limitation due to continued logging, or arbitrary signals associated with activity performing [2], they have proven to be effective in HAR.

B. EXTRACTION OF FEATURES IN EXISTING HAR STUDIES

The identification of the optimal feature set for HAR poses a considerable challenge. In the context of supervised learning, the extraction of features is instrumental in defining the fundamental attributes and characteristics underlying each segmented window, a crucial phase that determines the essential attributes in machine learning. The reduction of feature dimensionality, as accomplished by feature selection, serves to remove irrelevant and redundant features while preserving the integrity of the signal. As detailed in [2], [3], and [41], the advantages of feature selection in machine learning are manifold, including enhanced learning performance, such as higher accuracy and lower computational costs, as well as increased model interpretability. By selecting only a small subset of relevant features and removing noisy, irrelevant, and redundant ones, feature selection helps to improve model interpretability. This, in turn, simplifies the model, making it more accessible to understand, as it focuses solely on the most critical features () [41]. It is crucial to ensure that the selected features provide a significant difference between one activity and another. The extraction of features can be performed in either the statistical or frequency domain. These values are measured empirically for each windowed signal and represent variable attributes of that window.

In addition in [2] and [3], one or more features are generally not sufficient to differentiate one activity from another. It is better to have a combination of a few sets of features. For instance, Trost et al. [42] formulated statistical features from the accelerometer signal attached to the wrist and hip to classify different types of activities. Furthermore, Gao et al. [43] proposed the analysis of general features based on the time domain, frequency domain, and heuristic. Guiry et al. [44] analyzed the accelerometer signal from both smartphone and chest sensor before evaluating the discriminant power using several classification mechanisms such as Artificial Neural Network, Decision Tree, and k-Nearest Neighbor. In contrast to supervised learning, Kwon et al. [45] and Biswas et al. [46] explored unsupervised learning in classifying activities. Chung et al. [47] used frequency domain features from a chest-mounted sensor to distinguish between rest, walking, and running. In [47], a rough hierarchical threshold-based technique was implemented on Signal Magnitude Area (SMA). Research in [18] synchronized the data collection of the accelerometer with recorded video, and the features of the data were extracted using discrete wavelet transform (DWT). Some earlier works did not even use a classification algorithm but rather employed empirically determined thresholds for classification [48], [49], [50].

The field of activity recognition pertaining to human behavior has produced notable outcomes in the extraction of features technique, which can be broadly categorized into two types: hand-crafted features [51] and self-learning features [52]. The former is designed based on domain knowledge and is computed from each segmented window, representing the characteristics of the activity. Hand-crafted features have demonstrated successful applications in human activity recognition due to their lightweight implementation and suitability for ubiquitous devices [53], [54]. In contrast, deep learning techniques have also been employed to automatically extract features for human activity recognition. What sets deep learning networks apart is their ability to learn features that are very deep and do not require prior domain knowledge [55].

The existing literature on Human Activity Recognition (HAR) sheds light on the challenges and approaches in feature studies and highlights several gaps in the research. One of the major challenges identified is the identification of the optimal set of features for HAR. Although the importance of feature selection for dimensionality reduction is emphasized, the existing studies do not explicitly address the methodologies for systematically determining the optimal feature set, considering variations in activities and environmental contexts. Additionally, there is a limited exploration of combined feature sets, and although the advantages of using a combination of feature sets were mentioned, there is a gap in the discussion of systematic methods for selecting and combining these features. Hence, it is necessary to explore how different sets of features from various domains (statistical and frequency) can be effectively combined to improve the discriminatory power between activities. Lastly, the existing studies lack exploration of feature fusion. Although the categorization of features into hand-crafted and self-learning, it does not extensively explore how these types of features can be fused. Investigating how domain knowledge-based hand-crafted features can complement automatically learned features from machine learning approaches could offer a more comprehensive feature representation.

In summary, it is apparent that there are gaps in the optimal feature set identification, exploration of combined feature sets, and feature fusion. Addressing these gaps would not only contribute to the advancement of HAR systems but also enhance the applicability of such systems in diverse application.

C. NEIGHBORHOOD COMPONENT ANALYSIS (NCA)

The Neighborhood Component Analysis (NCA) is a highly effective feature selection method that identifies intrinsic properties, such as variance or correlation with the target variable, to determine the most significant features. Unlike other methods, NCA does not rely on the performance of a learning algorithm on the selected features. Instead,

it leverages the concept of local neighborhoods to determine an optimal linear transformation of data. This approach aims to enhance the performance of classification tasks by optimizing the feature space [56], ensuring that data points belonging to the same class are closer to each other, while those from different classes are pushed farther apart. The technique explicitly considers local neighborhood relationships, which enhances the discriminative power of the features [56]. NCA can handle multiclass problems by considering pairwise relationships between classes, and it is a non-parametric approach that does not assume any specific functional form for the decision boundaries. However, NCA can present computational challenges for large datasets and high-dimensional data due to the curse of dimensionality. In summary, NCA seeks to find an optimal linear transformation of the data that maximizes the discriminative power of the features [56]. By considering local neighborhood relationships, NCA aims to preserve or enhance the relationships between data points in a lower-dimensional space, ultimately improving the performance of classification tasks. Overall, NCA is a highly effective and sophisticated technique that can significantly enhance the performance of feature selection and classification tasks.

Numerous studies [57], [58], [59], [60], [61] have been conducted on the effectiveness of Neighborhood Component Analysis (NCA) for classification, which have demonstrated its ability to significantly improve nearest-neighbor-based classification tasks by learning an optimized distance metric. Goldberger et al. introduced NCA as a powerful technique for dimensionality reduction and feature selection tailored for k-nearest neighbor classifiers [57]. Chen et al. extended NCA by proposing a semi-supervised variant that addresses scenarios with limited labeled data [58]. Li et al. further investigated the application of NCA in deep learning, demonstrating its value in enhancing classification performance. Together, these studies underscore the ability of NCA to capture local relationships and improve the separability of classes, making it a valuable tool in solving various classification challenges. In another study, NCA was used in conjunction with deep learning techniques, specifically Linear Support Vector Machines (SVMs) and deep neural network (DNN), to enhance the performance of SVMs by learning optimized distance metrics [59]. The algorithm of NCA can be written as in work by Yang et al. [62], which multiclass, S consisted of set of feature vector, x_i and the y_i are the class labels denoted as

$$S = \{(x_i, y_i), i = 1, 2 \ldots, n\} \tag{1}$$

The computation of generalization error using neighborhood component analysis (NCA) was necessary to observe the improvement and determine if feature selection NCA need be done. The generalization error was implemented using none-fitting based method, L1 and fitting based method, L2. Assume L2 has higher value compared to L1, the reference point must be chosen randomly from $\boldsymbol{S_{train}}$. The probability of choosing the random reference point is denoted as

$$\mathrm{P(Ref(x)} = x_j \mid S) \propto kd_w(x_i, x_j) \tag{2}$$

The reference point for x is chosen from S, so sum of $P(\mathrm{Ref}(x) = x_j \mid S$ for all j must be equal to 1. Therefore, it is possible to write.

$$p_{ij} = \frac{kd_w(x_i, x_j)}{\sum_{j=1}^{n} kd_w(x_i, x_j)} \tag{3}$$

If and only if $i \neq$ j, then $p_{ij} = 0$ if and only if $i = j$. Meanwhile, k is some kernel or a similarity function that assumes large values when $d_w(f_{itrain}, f_{jtrain})$ is small. As k is written

$$k(z) = \exp(\frac{z}{\sigma}) \tag{4}$$

The kernel width σ is an input parameter that influences the probability of each point being selected as the reference point. As if $\sigma \to 0$, only the nearest neighbor of the query sample can be selected as its reference point. On the other hand, if $\sigma \to +\infty$, all the points have the same chance being selected apart from the query point. The reference point, x was unknown and selected randomly from $\boldsymbol{S}$. Meanwhile, the weighted distance, d_w between x_i and x_j is denoted as

$$d_w(x_i, x_j) = \sum_{n=1}^{r} w_l^2 |x_i - x_j| \tag{5}$$

where w_l is a feature weight that associated with lth feature. According to work by Yang et. al, the classification of nearest neighbour can be succeed, if and only if the classification of leave-one-out classification for matrix $\boldsymbol{S}$ is maximized [62]. The probability, p_{ij} of x_j being chosen as a reference point for new point x is higher depending on the closeness of the distance between the two samples as

$$p_i = \sum_{j}^{n} y_{ij} p_{ij} \tag{6}$$

where $y_{ij} = 1$ *if and only if* $y_i = y_j$ *and* $y_{ij} = 0$. Work in [62] has maximized the average leave-one-out classification accuracy can be written as

$$\xi(w) = \frac{1}{N} \sum_i p_i = \frac{1}{N} \sum_i \sum_j^n y_{ij} p_{ij} \tag{7}$$

Meanwhile, regularized neighborhood component analysis (RNCA) is a variation of NCA that enhances the NCA objective generalization function capabilities. Yang et al. has introduced the regularization terms, RNCA controls the complexity of the learned projection matrix and prevents overfitting by discouraging excessively large weight values [62]. The proposed method, which is called NCFS, uses the gradient ascent technique to maximize the expected leave-one-out classification accuracy with a regularization term [62]. The effectiveness of this algorithm has been evaluated through several experiments involving toy data and eight microarray datasets [62]. The proposed methods insensitive to a specific regularization parameter as shown

in Eq. 2.11 [62]. Regularization also penalizes irrelevant or redundant features, preventing RNCA from selecting overfitting features. RNCA has shown superior effectiveness over other feature selection methods, such as L1 and L2 regularization.

$$\xi(w) = \frac{1}{N} \sum_i \sum_j^n y_{ij} p_{ij} - \lambda \sum_{l=1}^d w_l^2 \tag{8}$$

The studies conducted by [63] suggest that RNCA is a regularization technique that effectively balances the trade-off between improving class separability and avoiding overemphasis on certain features. This technique is particularly useful in high-dimensional datasets where the risk of overfitting is significantly higher. RNCA provides a more controlled approach to dimensionality reduction with NCA, ultimately leading to better generalization performance in various classification tasks. Zhang et al. [63] have proposed a new feature selection method known as Normalized Neighborhood Component Feature Selection and Feasible-Improved Weight Allocation (NNCFS-FIWA). This method combines the strengths of neighborhood component analysis (NCA) and regularization to select features that are both relevant and informative. The NNCFS-FIWA method normalizes the data to make it more robust to noise [63], and then uses NCA to find a transformation of the feature space that minimizes the distance between data points of the same class.

Malan and Sharma have introduced a novel feature selection technique using regularized neighborhood component analysis (RNCA) to enhance the classification performance of motor imagery (MI) signals [64]. In [64] concluded that RNCA outperforms other feature selection algorithms, such as ReliefF, PCA, and GA, in terms of classification accuracy. Additionally, the methodology has proven to be effective in reducing the feature space, thereby leading to faster training and testing times for the prediction model when compared to other algorithms [64]. Moreover, [65] proposed a novel hybrid model using neighborhood components analysis, genetic algorithm, and support vector machine classifier (NCA-GA-SVM). The algorithm achieved the highest accuracy and can be successfully applied for the assessment of hepatocellular carcinoma mortality in the future [65]. On the hand, Yang and Laaksonen proposed Regularized Neighborhood Component Analysis (RNCA) as a solution to overfitting in Neighborhood Component Analysis (NCA) and Relevant Component Analysis (RCA) methods [66]. The study highlighted that NCA and RCA are effective for learning a low-dimensional linear embedding of labeled data by maximizing class prediction through neighboring samples in the transformed space [66]. However, these methods are prone to overfitting, particularly in high-dimensional data, which requires regularization [66]. The results of the study indicate that NCA and RCA may fail for high-dimensional data, resulting in heavily mixed classes in testing data and poor discrimination in transformed spaces, particularly in facial image processing [66]. On the other hand, RNCA preserves separation between classes in both training and testing data, thus alleviating the overfitting effect [66]. The regularization technique in RNCA improves the performance of neighborhood-based metrics, making it more suitable for discriminative information visualization and classification tasks [66].

Also, as per the results presented in reference [67], the RNCA-based approach has shown to outperform other feature selection techniques. Notably, the method has achieved an accuracy rate of over 96%, along with a recall rate exceeding 95% [67]. The study has placed emphasis on the importance of preserving context information of opcode, which was successfully accomplished by utilizing non-fixed length sequence fragments integrated with RNCA [67]. RNCA is a feature selection method that leverages the benefits of NCA and regularization techniques. NCA, a non-parametric method, selects features by transforming the feature space to minimize the distance between data points of the same class. On the other hand, regularization is applied to prevent overfitting and improve the generalization performance of machine learning models. RNCA offers several advantages over other feature selection methods, including improved nearest neighbor classification [57], the ability to select relevant features even in high-dimensional data [63], robustness to noise [63] and outliers, computational efficiency that makes it suitable for large datasets, and the potential to enhance the performance of machine learning models such as support vector machines, random forests, and logistic regression. Additionally, it can enhance multiclass handling [57].

NCA algorithm has several noteworthy limitations that must be considered. One of these limitations is NCA is sensitive to the choice of distance metric [57]. The nature of the distance metric utilized to gauge the similarity between data points can significantly impact the performance of NCA, with different metrics producing varying outcomes. As a result, the efficacy of NCA in maintaining intrinsic neighborhood relationships may be compromised [57]. Another limiting factor of NCA is that it is a linear technique, meaning it learns linear transformations of the feature space. This technique may not always capture complex non-linear relationships present in the data, particularly when the underlying data relationships are highly non-linear [68]. This limitation has prompted the development of non-linear variations and extensions of NCA to handle such scenarios [65].

D. MAXIMUM RELEVANCE AND MINIMUM REDUNDANCY (MRMR)

RNCA is a promising feature dimensionality reduction technique, but several research gaps warrant further exploration. Optimal strategies for regularization and hyperparameter tuning within the RNCA framework are essential, as they can significantly influence the quality of the learned representation. Additionally, scalability challenges when applied to large datasets need to be addressed, requiring the development of efficient RNCA algorithms. The extension

of RNCA to handle heterogeneous data types, such as categorical and continuous features, remains an open research area. Moreover, while RNCA can yield lower-dimensional representations, ensuring their interpretability is vital for real-world applications. By addressing these research gaps, RNCA can further empower with maximizing relevant and minimizing redundant (MRMR) as a valuable pre-feature filter in the field of feature dimensionality reduction. The MRMR algorithm finds an optimal set of features that is mutually and maximally dissimilar and can represent the response variable effectively. The algorithm minimizes the redundancy of a feature set and maximizes the relevance of a feature set to the response variable. Following is the equation of MRMR algorithm that quantifies the redundancy and relevance using the mutual information of variables—pairwise mutual information of features and mutual information of a feature and the response as

$$I(X,Z) = \sum_{i,j} P(X = x_i, Z = z_j) log \frac{P(X = x_i, Z = z_j)}{P(X = x_i, Z = z_j)} \tag{9}$$

where I is a mutual information of the discrete random variables X and Z. The goal of the MRMR algorithm is to find an optimal set S of features that maximizes V_S, the relevance of S with respect to a response variable y, and minimizes W_S, the redundancy of S, where V_S and W_S are defined with mutual information I

$$V_S = \frac{1}{[S]} \sum_{x \in S} I(x, y) \tag{10}$$

$$W_S = \frac{1}{[S]\,2} \sum_{x \in S} I(x, z) \tag{11}$$

|S| is the number of features in S. Finding an optimal set S requires considering all $2^{|\Omega|}$ combinations, where Ω is the entire feature set. Instead, the MRMR algorithm ranks features through the forward addition scheme, which requires $O(|\Omega| \cdot |S|)$ computations, by using the mutual information quotient (MIQ) value

$$MIQx = \frac{Vx}{Wx}. \tag{12}$$

where Vx and W_x are the relevance and redundancy of a feature, respectively.

$$Vx = I(x, y) \tag{13}$$

$$W_S = \frac{1}{[S]} \sum_{x \in S} I(x, z) \tag{14}$$

The Minimum Redundancy Maximum Relevance (MRMR) feature selection method is highly advantageous in the realm of machine learning and data analysis. MRMR offers several key benefits, making it a valuable tool for data scientists and researchers. It aids in improving model performance by selecting features that are highly relevant to the target variable, thereby reducing noise and enhancing prediction accuracy [69]. Additionally, MRMR effectively tackles the issue of high-dimensional data by reducing the number of features to a more manageable subset, leading to more efficient model training and reduced overfitting [69], [70]. This method also enhances interpretability where understanding the relationship between variables and outcomes is essential [71]. Moreover, MRMR contributes to faster model training and inference, reduces sensitivity to noisy data, and encourages robust generalization [71]. Its emphasis on relevance and the reduction of redundancy ensures that models are better at making predictions on new, unseen data while mitigating the risk of overfitting [69], [70], [71]

The Neighborhood Component Analysis (NCA) is a highly effective feature selection method that identifies intrinsic properties, such as variance or correlation with.

III. METHOD

A. DATASETS ACQUISITION

The study employed two separate datasets to achieve accurate recognition results. The first dataset, referred to as FHAR, was specifically designed to recognize field hockey activities. The second dataset, which was already in existence, served as the basis for human recognition.

- The Field Hockey Activity Recognition (FHAR): the study employed commercially available Physilog®5 motion sensors, which feature high-quality 3D accelerometer, 3D gyroscope, and barometric pressure sensors with programmable sampling frequency and range. The sensors continuously captured accelerometer and gyroscope readings at a frequency of 128Hz. Four inertial sensors were affixed to the chest, waist, right, and left wrist of each of the 43 state-level field hockey athletes using Velcro straps. During the study, the participants performed six fundamental field hockey activities: passing, driving, drag flick, dribbling, receiving, and tackling. The activities were to repeat each activity three times, generating signal data consisting of three repetitions of passing, three repetitions of driving, three repetitions of drag flick, five longer and continuous dribbling, three repetitions of receiving, and three repetitions of tackling. To increase the number of samples, this cycle was repeated three times. In this study, the combination of sensors such as two, three and all four sensors were considered and studied.
- Human activity recognition (HAR1) dataset: The study conducted by Huang and Onnela in 2020 [72] involved the collection of raw gyroscope and accelerometer data utilizing various techniques for carrying the smartphone. These methods encompassed placing two smartphones in the front and back pants pockets, respectively, and reorienting the front pocket phone in the four possible directions. Apart from data from smartphones, information from wearable devices was also gathered. The research involved four healthy subjects, including two females and two males, who met the eligibility criteria of being at least 18 years old and capable of walking, standing, and

ascending and descending stairs unaided. For each participant, age, weight, height, sex, dominant hand, and the preferred method of carrying their phone were recorded.

B. SEGMENTATION

The FHAR signal from the datasets was segmented into sliding windows of varying sizes to analyze human activities through sensor data where $\boldsymbol{x}_j = \{x_i\} \, | i = \{1, 2, 3 \ldots \mathrm{N}\}, j = \{1, 2, 3 \ldots \mathrm{D}\}$. The window size was determined by N, which represents the number of samples in the sliding window, and D, which represents the total number of signals from 4 sensors, with 6 signals (3 from acceleration and 3 from gyroscope signals) per sensor. The sliding window sizes were evaluated for different sensor configurations, including two, three, and all four sensors, to measure the impact of sensor configuration on the accuracy of the analysis.

For each sensor configuration, approximately three window sizes were evaluated, which were (N=179 samples/window), (N=115 samples/window), and (N=141 samples/window), respectively. The sliding window size of 179 samples/window was used for the configuration of two sensors, the sliding window size of 115 samples/window was used for the configuration of three sensors, and the sliding window size of 141 samples/window was used for the configuration of all four sensors. This allowed for a comprehensive analysis of the impact of sensor configuration and window size on the accuracy of the analysis.

For the HAR1 datasets, the signal was segmented into N=30 samples/window to analyze human activities through sensor data. Rectangular windows were used for each, with an overlap of 50% of its size. Overlapping of the window is necessary to handle transitions more accurately, as sports movements are extremely vigorous and physically rugged. This approach allowed for a more detailed analysis of the data, enabling researchers to better understand the dynamics of human activities and improve the accuracy of the analysis.

C. FEATURE EXTRACTION

The models that performed the best were further improved by incorporating 18 additional features. These features were obtained from other time and frequency domain features and were used as inputs to the classification algorithm. The 18 features included mean ($f_{\mu}^{[j]}$), standard deviation ($f_{\sigma}^{[j]}$), minimum ($f_{max}^{[j]}$) and maximum ($f_{min}^{[j]}$) variance ($f_{v}^{[j]}$), median $f_{median}^{[j]}$, mode ($f_{mode}^{[j]}$), range ($f_{range}^{[j]}$), kurtosis ($f_{kurtosis}^{[j]}$), skewness ($f_{skw}^{[j]}$), harmonic mean ($f_{m}^{[j]}$), root mean square ($f_{rms}^{[j]}$), interquartile ($f_{IQR}^{[j]}$), mean absolute deviation ($f_{MAD}^{[j]}$), band power ($f_{MNP}^{[j]}$), mean frequency ($f_{MNF}^{[j]}$), median frequency ($f_{MDF}^{[j]}$) and power bandwidth ($f_{BW}^{[j]}$) as shown in (15) - (32). The equations (15) through (32) clearly illustrate these features. By including these additional features, the models were able to provide more detailed and accurate results.

$$f_{\mu}^{[j]} = \frac{1}{N} \sum_{i=1}^{N} x_j(i) \tag{15}$$

$$f_{\sigma}^{[j]} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} x_j(i) - ^2} \tag{16}$$

$$f_{max}^{[j]} = max x_j(i) \tag{17}$$

$$f_{min}^{[j]} = min x_j(i) \tag{18}$$

$$f_{v}^{[j]} = \frac{1}{N-1} \sum_{i=1}^{N} x_j(i) - ^2 \tag{19}$$

$$f_{median}^{[j]} = \boldsymbol{x_j}\,(i)\,[\left(\frac{N+1}{2}\right)] \tag{20}$$

$$f_{MDF}^{[j]} \textit{ represented by the most frequent value} \tag{21}$$

$$f_{range}^{[j]} = f_{max}^{[j]} - f_{min}^{[j]} \tag{22}$$

$$f_{kurtosis}^{[j]} = \frac{1}{N} \sum_{i=1}^{N} \frac{(\boldsymbol{x_j}(i) - \mu)^4}{\sigma^4} \tag{23}$$

$$f_{skw}^{[j]} = \frac{1}{N} \sum_{i=1}^{N} \frac{(\boldsymbol{x_j}(i) - \mu)^3}{\sigma^3} \tag{24}$$

$$f_{m}^{[j]} = \frac{N}{\sum_{i=1}^{N} \frac{1}{\boldsymbol{x_j}(i)}} \tag{25}$$

$$f_{rms}^{[j]} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} [\boldsymbol{x_j}(i)]^2} \tag{26}$$

$$f_{IQR}^{[j]} = Q_3 - Q_1 \tag{27}$$

$$f_{MAD}^{[j]} = \frac{\sum_{i=1}^{N} |\boldsymbol{x_j}(i) - \mu|}{N} \tag{28}$$

$$f_{MNP}^{[j]} = \sum_{i=1}^{N} \frac{\boldsymbol{x_j}(i)}{N} \tag{29}$$

$$f_{MNF}^{[j]} = \frac{\sum_{i=1}^{N} \boldsymbol{fx}}{\sum_{i=1}^{N} \boldsymbol{f}} \tag{30}$$

$$f_{MDF}^{[j]} = \frac{1}{2} \sum_{i=1}^{N} \boldsymbol{x_j}(i) \tag{31}$$

$$f_{BW}^{[j]} \textit{ represented by the frequency range} \tag{32}$$

D. FEATURE SELECTION

At the end, there are 108 features per each sensor (D × 18 types of features) were used at this stage. The workflow of the integration of relevant and irrelevant features before applying the RNCA algorithm onto the feature-class matrix, assuming, $\boldsymbol{S}$. The relevance score of each feature in $\boldsymbol{S}$ was calculated using (13) concerning the target class, indicating the predictive power of each feature individually. The redundancy score between each pair of features was computed using (14), which measures the similarity or correlation between two features and their degree of redundancy. Pearson correlation coefficient was employed as a commonly used measure of redundancy. The highest relevant score feature was identified and added to the new feature-class, $\boldsymbol{S_1}$. For each remaining feature, the relevance score and similarity score were computed in the subsequent iteration. In the new $\boldsymbol{S_1}$ the features were ranked based on relevance, with lower priority given to redundancy.

To evaluate the selected features, the performance of the selected feature subset was assessed and compared with the former $\boldsymbol{S}$ features using a t-test. This evaluation generates two hypotheses, H0 and H1, referred to as the null hypothesis and the alternative hypothesis, respectively. H0 assumes that there is no significant loss of accuracy between the two accuracies being compared. H1 asserts that there is a significant loss of accuracy between the two accuracies being compared. The p-value in a t-test represents the probability of obtaining the observed sample data or more extreme data. If the p-value is greater than or equal to the significance level (α), which is 0.05 (or 5%), it is not statistically significant. In this case, the null hypothesis is not rejected.

The selection of features was computed iteratively only if the p-value was less than a predetermined significance level (α), which was considered statistically significant. A model-selected feature supported by t-test (H0 and p-value $\approx$ 1) is considered as the best feature combination with reducing feature dimension to proceed with the RNCA algorithm, known as a new multi-class, $\boldsymbol{S_1}$. The new configuration of $\boldsymbol{S_1}$ was split into a training set and a testing set with a ratio of 0.95:0.05. Two generalization errors (L1 and L2) were computed to measure the necessity of the NCA algorithm upon the new S. L1 is the error rate of a model without using the NCA algorithm on S. Empirically, L1 regularization is applied to a model without undergoing the training or fitting process with data. This study applies L1 regularization directly to parameters without fitting the model to observe its status. However, this method may limit the generalization capability of the model, as it does not take new data patterns into account.

On the other hand, L2 is the error rate of the model with fitting the NCA algorithm, where the value of λ is set to zero. When λ is set to zero, the L2 regularization term essentially disappears from the objective function, and the model is trained without any penalty for large weights. This means that the optimization process focuses solely on minimizing the original loss, and there is no additional penalty for having large coefficients. If L1 $>$ L2, the model requires NCA. A 10-fold cross-validation with regularization of λ was computed on the training set. In l0-fold cross-validation, different values of λ were evaluated by training and testing the model on different subsets of the data. A minimum loss of MSE was generated that corresponds to the optimal value of λ that achieves the best tradeoff between model complexity and generalization performance.

Feature weights based on the best λ were analyzed. The higher the feature weight, the more important the feature. The evaluation of features was studied on different levels of feature weight, such as feature weights above 0 and feature weights above 1. The accuracies of each level were compared and observed. The RNCA was able to differentiate the features based on the feature weight. The proposed approach leads to two hypotheses: 1) same accuracy with reduced feature dimension and 2) higher accuracy with reduced feature dimension. The sensor configuration, performance classification, and feature dimension were observed and compared.

IV. RESULTS AND DISCUSSION

A. FEATURE SELECTION BY MRMR

Fig. 1a)-d) depict the feature ranking of MRMR using bar plots, where the x-axis represents the features, and the y-axis portrays the ranking scores calculated using a combination of relevance and redundancy metrics. Each feature's height in the plot corresponds to its ranking score, indicating its significance for classification purposes. Conversely, shorter bar features may be considered redundant or less significant. The score difference between consecutive bars is noteworthy, implying a small difference between features. The p-value for each model obtained from repeated cross-validation tests was approximately 1, failing to reject the null hypothesis. This suggests that the MRMR method does not significantly affect the performance of the decision tree, and the observed difference in accuracies may be due to random chance.

The algorithm MRMR with a 10-by-10 repeated cross-validation t-test was used for feature selection. The algorithm aimed to select the most informative subset of features by considering both the relevance of features to the target class label and their redundancy with each other. The use of repeated cross-validation in the evaluation process enhanced the reliability and robustness of the results by accounting for the potential variability in the data splits and reducing the impact of random partitioning, which provided a more comprehensive assessment of the performance by averaging the results across multiple iterations. Overall, the application of MRMR with a 10-by-10 repeated cross-validation t-test demonstrated a systematic approach to feature selection while considering both relevance and redundancy before the NCA algorithm.

B. REGULARIZATION PARAMETER VERSUS MEAN SQUARED ERROR (MSE)

Fig. 2a)-d) provides a detailed comparison of different sensor configurations for two distinct models, FHAR and HAR1. The Fig. 2a)-d) displays their best regularization parameters and corresponding losses, with the best lambda values achieved for each configuration being 0.00058, 0.00012, and 0.00015, respectively. Additionally, the associated losses for the two-, three-, and all four-sensor setups are recorded as 0.0795, 0.065, and 0.0628. It is noteworthy that using all four sensors leads to the lowest loss, suggesting a potentially more optimal sensor arrangement for the FHAR model. However, the HAR1 model's entry only displays results for the two-sensor configuration, with the best lambda achieved being 0.0284, significantly higher than any FHAR configuration, and the corresponding loss being 0.1071 as shown in Fig. 2d. This variation in best lambda and loss metrics across sensor configurations in both models highlights the importance of selecting the right combination of sensors for optimizing

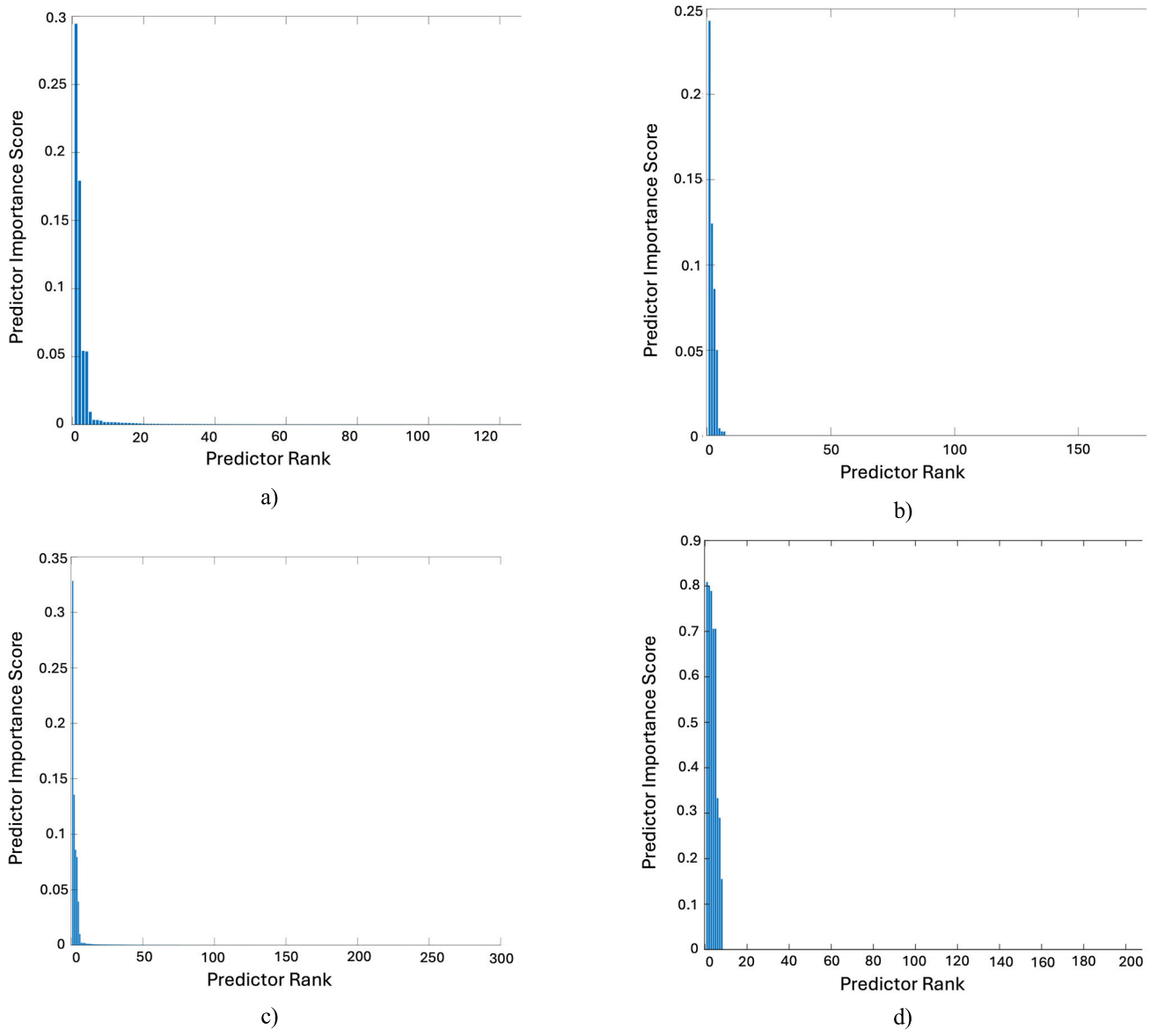


FIGURE 1. Feature ranking by MRMR for a) two sensors, b) three sensors, c) all four sensors of FHAR data, d) HAR1 dataset.

model performance. It is crucial to identify the optimal combination to achieve the best performance in each case, as different models might respond uniquely to various sensor setups.

The loss mentioned in Fig. 2a)-d) represents the objective function or loss function value associated with the specific lambda value found through the RNCA algorithm. In the context of RNCA, the loss function could represent a measure of error or discrepancy between the predicted outputs and the actual targets in the training data. As a general rule, lower loss values indicate better performance. Different lambda values have been tried with various sensor configurations. The combination of lambda and the corresponding loss helps in assessing the trade-off between model complexity (controlled by lambda through regularization) and predictive performance. In the context of RNCA, selecting the best lambda involves finding a balance between model complexity and fitting the training data well. This balance ensures that the model generalizes effectively to new, unseen data while avoiding overfitting. The specific lambda and its associated loss value in the table represent the optimal configuration found through the RNCA process for each sensor setup in the given models (FHAR and HAR1).

C. FULL FEATURES VERSUS FEATURES BASED MRMR VERSUS FEATURES BASED MRMR_RNCA

Fig. 3 showcases the feature counts resulting from different feature selection methods applied to two distinct models, FHAR and HAR1, each with varying sensor configurations. The objective of the feature selection process was to curate

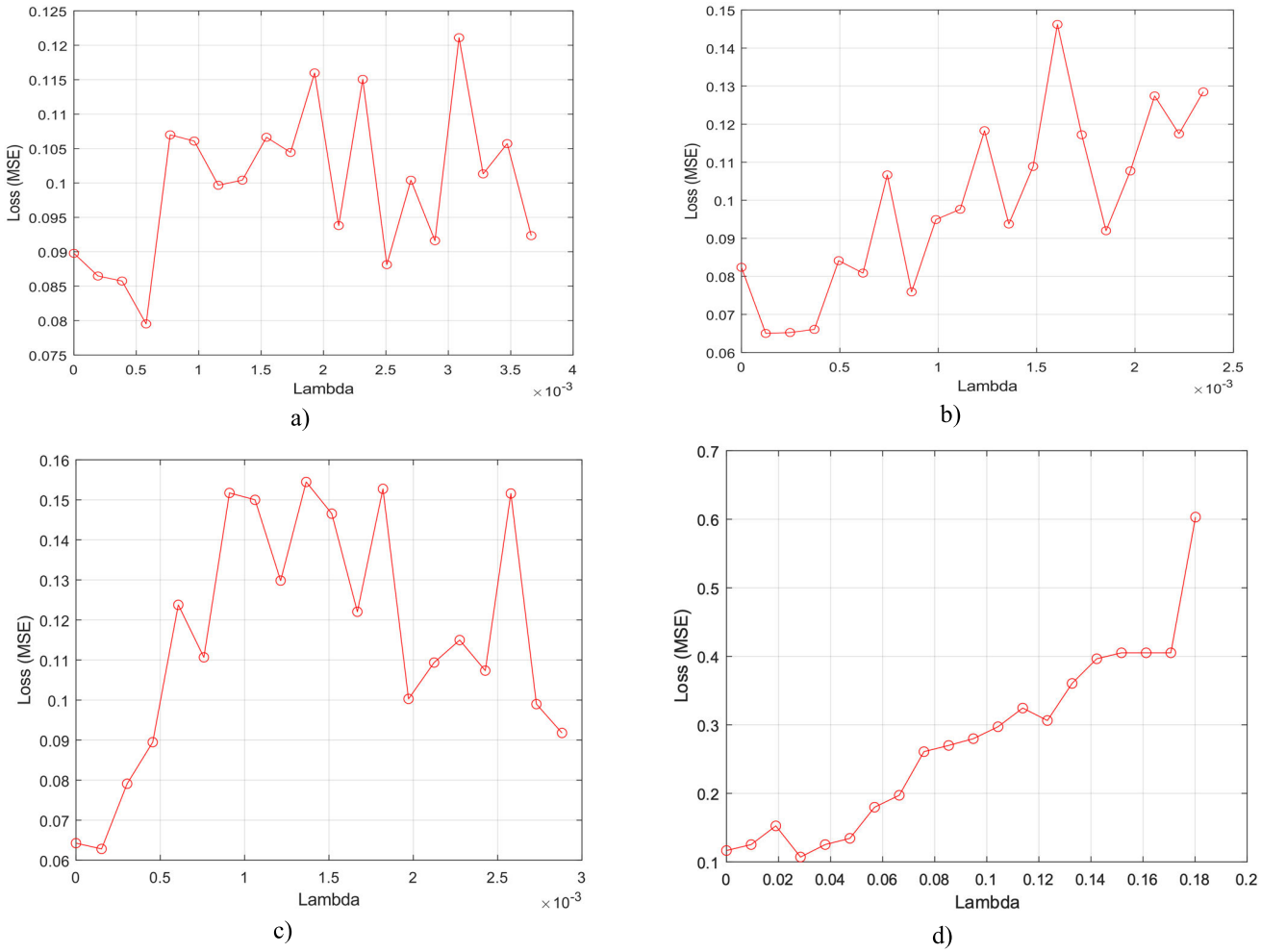


FIGURE 2. Regularization of λ for a) two sensors, b) three sensors, c) all four sensors of FHAR data, d) HAR1 dataset.

the most informative features while minimizing redundancy. The report delves into more specific subsets of feature selection, particularly involving MRMR with RNCA (Regularized Neighborhood Component Analysis) constraints. The analysis highlights the pivotal role of feature selection strategies, particularly those integrating RNCA, in determining the subset of features deemed most relevant and non-redundant for predictive modeling within specific sensor configurations and models. At the outset, the ''Full features'' column serves to provide a reference for the total number of available features without any selection. Moving to ''Features Based on Maximum Relevance Minimum Redundancy (MRMR),'' it becomes evident that this method aims to reduce redundancy while selecting the most informative features. The subsequent columns, ''Features Based MRMR_RNCA above 0'' and ''Features Based MRMR_RNCA above 1,'' offer deeper insights into the impact of Regularized Neighborhood Component Analysis (RNCA) on feature selection, filtering features based on their RNCA scores.

For the FHAR model, the feature counts vary notably among different sensor configurations (two, three, and all four sensors) and selection methods. When all four sensors are utilized, the counts differ more across the methods, particularly in the ''Features Based MRMR_RNCA above 0'' category. This variance suggests that the feature selection process, especially when integrating RNCA scores, significantly influences the number of selected features, potentially impacting model complexity and performance. In contrast, for the HAR1 model, although feature counts differ across selection methods, the differences are relatively smaller than those in FHAR, implying a different nature of feature relevance and redundancy within this particular model's context. The disparities in feature counts across selection methods and sensor configurations underscore the significance of the chosen feature selection approach and its interplay with the specific model and sensor setup in optimizing the model's performance and computational efficiency. This report emphasizes the pivotal role of feature

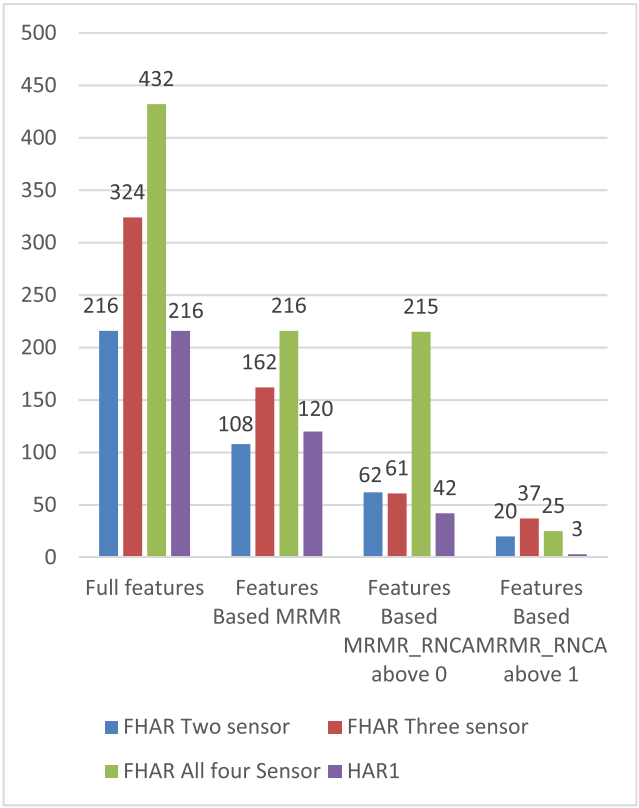


FIGURE 3. Features combination resulted from different features selection approaches.

TABLE 1. Comparison accuracies of different feature counts.

	FHAR			HAR1
	Two sensors	Three sensors	All four sensors	
Full features	92.8%	93.2%	94.4%	88.3%
Features Based MRMR	93.1%	93.4%	94.2%	88.9%
Features Based MRMR_RNCA above 0	93.5%	93.7%	94.5%	90.1%
Features Based MRMR_RNCA above 1	90.0%	92.4%	91.0%	76.3%

selection strategies, particularly those integrating RNCA, in determining the subset of features deemed most relevant and non-redundant for predictive modeling within specific sensor configurations and models.

D. PERFORMANCE OF ACCURACY

Table 1 showcases the classification accuracies of two different models, FHAR and HAR1, under various sensor configurations using different feature selection methods. Starting with a baseline represented by "Full Features," indicating the use of all available features, subsequent rows depict the impact of specific feature selection methodologies on classification accuracy. "Features Based on Maximum Relevance Minimum Redundancy (MRMR)" displays slight improvements across sensor configurations, highlighting the method's effectiveness in selecting more informative features. However, as Table 1 progresses to "Features Based on MRMR with RNCA Constraints," a nuanced pattern emerges. The "Features Based MRMR_RNCA above 0" consistently outperforms the MRMR approach, indicating the added benefit of integrating Regularized Neighborhood Component Analysis (RNCA) scores in feature selection, enhancing accuracy across both models and sensor setups. Nevertheless, the strictest constraint, "Features Based MRMR_RNCA above 1," leads to a notable accuracy decline, particularly visible in the FHAR model across all sensor configurations and notably affecting the HAR1 model with two sensors. This suggests that while stringent RNCA-based constraints might yield smaller feature subsets, they don't uniformly guarantee accuracy improvements, highlighting the delicate balance required in selecting feature subsets for optimal model performance across varying sensor configurations.

V. CONCLUSION

Activity recognition is a complex task that can be approached in various ways. One of the commonly used methods involves the use of wearable sensors to track player movements and other physiological data, which can then be used to estimate the activity they are performing. In field hockey, activity recognition plays a crucial role for notational analysts. However, developing automated activity recognition systems can be challenging due to the different modality configurations and frameworks, resulting in issues related to data processing, flexibility, and robustness. This study aimed to provide a comprehensive explanation of a machine learning-based framework that utilizes wearable-based modality for activity recognition in field hockey. The study involved validating the models through experimental and comparative work and selecting the optimal window sliding size based on a variety of sliding window ranges. The results of the study showed that different sensor configurations have different optimal sizes of window sliding concerning the best accuracy and best F1 score. The most effective sensor configuration included all four sensors, which recorded the highest accuracy and F1 score compared to other sensor configurations.

Onto that, increasing the number of inertial sensors in a system typically leads to the establishment of a wider range of features and more comprehensive insights about activities and orientation. Incorporating multiple sensors results in an expanded dataset. This expanded dataset allows for more in-depth analysis and feature extraction on different statistical and frequency-based features that yielded 108 features per sensor. Dealing with high feature dimensions resulting from data collected from multiple sensors presents several challenges. The "curse of dimensionality" becomes a prominent

concern, as the exponential growth in data volume can overwhelm computational resources and storage. Also, data complexity escalates, making it harder to discern patterns and relationships within the data. Noise from irrelevant or redundant features can also negatively impact data analysis and modelling which leads to interpreting and visualizing high-dimensional data becoming more intricate.

Next, a valuable dimensionality reduction and feature selection technique, RNCA is especially suited for addressing the challenges associated with small sample sizes (single sensor) and high feature dimensions (two, three, and all four sensors). This method effectively narrows down the feature space by selecting a subset of features that are most informative for explaining the variance and structure of the data. The findings that the regularization has imposed a penalty on the transformation process, encouraging sparsity in the selected features only for small feature dimensions, chest sensor, (C). This sparsity constraint was especially useful in scenarios where traditional approaches might lead to overfitting. Conversely, in scenarios where the high feature dimension is significant such as in sensor configuration of two, three, and all four sensors, RNCA offers several disadvantages. The findings show that RNCA inefficiently reduces the dimensionality as the regularization parameter of 0 means no regularization was applied. In other words, overfitting occurs when the model becomes too complex or flexible, and it tries to fit the training data too closely.

Based on the preceding predicted results, an improved design was then proposed, where the feature selection was made using the integration of two feature selections, MRMR and RNCA-based framework. It consisted of two stages of feature selection: maximizing relevance and removing redundancy and constraining the feature with a neighbor-based approach. The proposed (MRMR and RNCA) were successfully analyzed and exhibited positive outcomes. It has improved accuracy by reducing unnecessary features. Still, among all the sensor configurations, all four sensors have shown the best modality for activity recognition in field hockey.

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